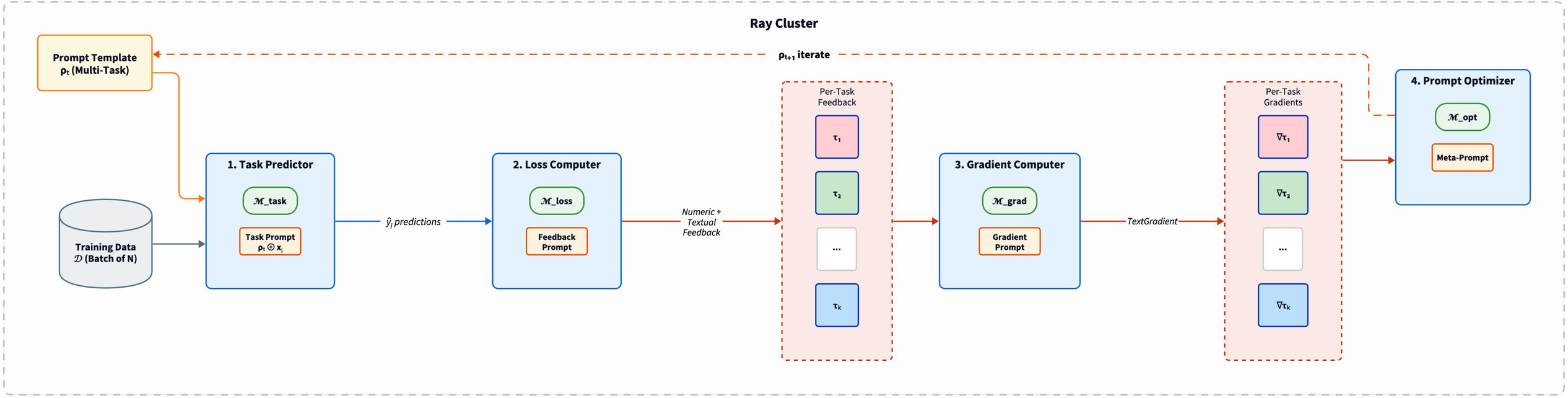
Ray Cluster
Prompt Template
ρ_t (Multi-Task)
ρ_{t+1} iterate
Training Data
$\mathcal{D}$ (Batch of N)
1. Task Predictor
$\mathcal{M}$_task
Task Prompt
$\rho_t \oplus x_i$
$\hat{y}_i$ predictions
2. Loss Computer
$\mathcal{M}$_loss
Feedback Prompt
Numeric + Textual Feedback
Per-Task Feedback
τ_1
τ_2
...
τ_k
3. Gradient Computer
$\mathcal{M}$_grad
Gradient Prompt
TextGradient
Per-Task Gradients
$\nabla\tau_1$
$\nabla\tau_2$
...
$\nabla\tau_k$
4. Prompt Optimizer
$\mathcal{M}$_opt
Meta-Prompt